FINANCIAL STATEMENT ANALYSIS · CASE SET

Context-Based Problems

DuPont Analysis and Altman Z-score

Six short cases with hypothetical data, each built around one learning point

STUDENT EDITION · problem and final answer

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Case	Title	Company	Learning point
D1	The Growth That Diluted Returns	Kaveri Appliances Ltd.	Leverage can mask weakening operations: split the change in ROE into its drivers (3-way and 5-way).
D2	Two Roads to the Same ROE	Aurum Jewels Ltd. and Bazaar Fresh Ltd.	Same ROE does not mean same risk: identify the ROE engine and test its sensitivity to a margin squeeze.
D3	The ROE That Nearly Doubled	Sunrise Textiles Ltd.	Headline ROE versus core ROE: strip out one-off gains and their tax effect before judging performance.
Z1	A Safe Borrower Slips	Deccan Auto Components Ltd.	Reading the zones: break the change in the Altman Z-score into its five weighted components.
Z2	When the Share Price Falls Faster than the Business	Marine Chemicals Ltd.	Market-value dependence: how much of Z is the share price, the break-even price, and a book-value cross-check (Z″).
Z3	No Share Price, No Original Z	Gokul Engineering Pvt. Ltd.	Choosing the right model: Z′ for private manufacturers, and why balance-sheet fixes move Z′ less than expected.

How to use this document

- Each case is a short context-based problem followed by the final answer so that you can check your own work.
- Work out the solution yourself first; the final answer gives the key results only.
- All companies, people and data are **hypothetical**. Amounts are in ₹ crore unless stated.

CASE D1 · DUPONT ANALYSIS

The Growth That Diluted Returns

Kaveri Appliances Ltd.

Learning point: Leverage can mask weakening operations: split the change in ROE into its drivers (3-way and 5-way).

Kaveri Appliances, a Pune-based maker of mixers, induction cooktops and water heaters, has just closed its best sales year ever. In FY2026 it opened a second plant at Nashik and added 300 dealers in tier-2 towns. At the results call, the managing director said: *"Profits are up, the business is bigger, and our shareholders are better off."*

Meera Joshi, an equity analyst at Sahyadri Securities, is not convinced. Revenue grew 20%, but profit rose only about 4%. The Nashik plant was funded largely by a new term loan. The board also kept the dividend high, so equity grew by only ₹50 crore. Before she publishes her note, her research head has asked her to explain what really happened to the return earned by shareholders.

The Nashik plant came on stream late in the year, so year-end assets are much higher than the assets that earned most of the year's revenue, and the new loan carries a higher interest rate than the older borrowings. She plans to use the DuPont framework, first in its simple three-way form and then in its fuller five-way form, so that the research head can see the reasons behind the change and not just the headline number.

₹ crore (balances at year-end)	FY2025	FY2026
Revenue	1,200.0	1,440.0
EBIT	168.0	187.2
Interest expense	28.0	41.2
Profit before tax (PBT)	140.0	146.0
Tax expense	35.0	36.5
Profit after tax (PAT)	105.0	109.5
Total assets	800.0	1,080.0
Shareholders' equity	400.0	450.0

Required

1. Compute the **three-way** DuPont ratios (net profit margin, asset turnover, equity multiplier) for both years and verify ROE.
2. Split the net profit margin further and compute the **five-way** ratios: tax burden, interest burden, EBIT margin, asset turnover and equity multiplier.
3. Show how many percentage points each driver added to, or took from, the change in ROE.
4. Evaluate the managing director's claim. Does ROE support it? Is leverage the explanation?
5. In two or three sentences, state what Meera should recommend.

Final answer

Measure	FY2025	FY2026
ROE	26.25%	24.33%
Net profit margin	8.75%	7.60%
Asset turnover	1.500×	1.333×
Equity multiplier	2.000×	2.400×
Tax burden	0.750	0.750
Interest burden	0.833	0.780
EBIT margin	14.00%	13.00%

ROE fell by 1.92 pp. Asset turnover (−2.98 pp), EBIT margin (−1.87 pp) and interest burden (−1.67 pp) pulled it down; the higher equity multiplier (+4.61 pp) partly offset them. The managing director's claim is not supported: return on assets fell from 13.1% to 10.1%, and the extra leverage only masked this.

CASE D2 · DUPONT ANALYSIS

Two Roads to the Same ROE

Aurum Jewels Ltd. and Bazaar Fresh Ltd.

Learning point: Same ROE does not mean same risk: identify the ROE engine and test its sensitivity to a margin squeeze.

Kiran Rao manages a ₹50 crore family-office portfolio and must choose one of two listed consumer companies for a three-year holding. Both reported an ROE of about 18% for FY2026, and a broker note concludes: *"Similar returns, so pick either."* Kiran is not so sure, and has asked an analyst to break each company's return into its parts.

Aurum Jewels sells branded gold and diamond jewellery through 60 large showrooms. Every piece carries a high mark-up, but stock moves slowly and the company is financed mostly by shareholders' money. **Bazaar Fresh** runs 400 neighbourhood supermarkets. Mark-ups are thin, shelves empty within days, and the stores are funded largely by supplier credit and bank loans.

Kiran's economist warns that wage and input-cost inflation could squeeze the operating margin of both businesses by about one percentage point of revenue next year. The two companies pay out a similar share of profit as dividends, so ROE is a fair basis for comparison. The family office's mandate is capital preservation first and growth second, so a company whose earnings are fragile in a bad year would be a poor fit even if it looks equally profitable today. Before committing, Kiran wants to know whether "same ROE" really means "same risk".

FY2026, ₹ crore (balances at year-end)	Aurum Jewels	Bazaar Fresh
Revenue	800.0	4,000.0
EBIT	168.0	100.0
Interest expense	8.0	30.0
Profit before tax (PBT)	160.0	70.0
Tax expense	40.0	17.5
Profit after tax (PAT)	120.0	52.5
Total assets	1,000.0	1,000.0
Shareholders' equity	650.0	285.0

Required

1. Compute the five DuPont ratios for both companies and verify ROE.
2. Classify each company's ROE engine as **margin-led, turnover-led or leverage-led**, giving reasons from the ratios.
3. Assume the EBIT margin falls by one percentage point of revenue, while revenue, interest, the tax rate, assets and equity stay unchanged. Recompute each company's ROE and interest cover (EBIT ÷ interest).
4. Which company would you choose for a risk-averse family office, and why?

Final answer

Measure	Aurum	Bazaar
ROE	18.46%	18.42%
Tax burden	0.750	0.750
Interest burden	0.952	0.700
EBIT margin	21.00%	2.50%
Asset turnover	0.800×	4.000×
Equity multiplier	1.538×	3.509×
ROE after 1 pp margin squeeze	17.54%	7.89%
Interest cover after squeeze	20.0×	2.0×

Aurum is margin-led with low leverage; Bazaar is turnover- and leverage-led on a 2.5% margin. After the squeeze Aurum's ROE falls 5% in relative terms, Bazaar's falls 57%. A risk-averse family office should choose Aurum.

CASE D3 · DUPONT ANALYSIS

The ROE That Nearly Doubled

Sunrise Textiles Ltd.

Learning point: Headline ROE versus core ROE: strip out one-off gains and their tax effect before judging performance.

Arjun Menon, a fund manager at Tarang Mutual Fund, runs a quarterly screen for companies with improving ROE. This quarter, Sunrise Textiles, a Coimbatore-based maker of cotton yarn and fabric, tops the list: ROE has risen from 12.0% to 20.6%. The press release calls FY2026 "a year of operational excellence".

Arjun reads the annual report and finds three things. First, cotton prices rose sharply and operating margins came under pressure; revenue grew only 3%. Second, the company sold a surplus plot of land for a gain of ₹55 crore, reported as an exceptional item within EBIT, and used the proceeds to repay loans, so total assets and interest both fell. Third, the gain was taxed at 20%, against 30% on regular business profit, so the effective tax rate fell.

Arjun remembers that the same screen once flagged a company whose profit was lifted by a one-off asset sale, and the stock later disappointed. Before adding Sunrise to the fund's buy list, the investment committee wants to know how much of the ROE improvement is real and how much comes from a one-time event.

₹ crore (balances at year-end)	FY2025	FY2026
Revenue	900.0	930.0
Operating profit before exceptional item	81.0	74.4
Exceptional gain on sale of land	0.0	55.0
EBIT (total)	81.0	129.4
Interest expense	21.0	19.4
Profit before tax (PBT)	60.0	110.0
Tax expense (FY2026 includes ₹11.0 on the land gain)	18.0	27.5
Profit after tax (PAT)	42.0	82.5
Total assets	700.0	660.0
Shareholders' equity	350.0	400.0

Required

1. Compute the five-way DuPont ratios for both years using reported figures (EBIT includes the exceptional gain).
2. Identify which ratios drove the rise in ROE and which worked against it.
3. Normalise FY2026: remove the exceptional gain and the tax on it, assume the regular 30% tax rate on the remaining profit, and compute the core EBIT margin, core PAT and core ROE (use closing equity as reported).
4. Compare core ROE with FY2025 ROE and write a three-line note to the committee.

Final answer

Measure	FY2025	FY2026
Reported ROE	12.00%	20.63%
Tax burden	0.700	0.750
Interest burden	0.741	0.850
EBIT margin	9.00%	13.91%
Asset turnover	1.286×	1.409×
Equity multiplier	2.000×	1.650×
Core EBIT margin	9.00%	8.00%
Core PAT (₹ crore)	42.0	38.5
Core ROE	12.00%	9.63%

Reported ROE rose 8.6 pp, but the gain explains the rise: core ROE fell from 12.0% to 9.6% (10.8% even if the retained gain is removed from equity). The underlying business weakened.

CASE Z1 · ALTMAN Z-SCORE

A Safe Borrower Slips

Deccan Auto Components Ltd.

Learning point: Reading the zones: break the change in the Altman Z-score into its five weighted components.

Anita Desai is a credit analyst at Mahanadi Bank. Every year she reviews the ₹150 crore working-capital limit extended to Deccan Auto Components, a supplier of brake and clutch parts to passenger-car makers. FY2026 was difficult: two large customers cut production schedules in the second half, sales fell 4%, finished-goods stock piled up, and the company drew more short-term loans to fund the inventory. The share price also fell as investors marked down the auto-ancillary sector. The promoters have asked for the limit to be renewed unchanged, and Anita knows that auto-component makers are capital-intensive and cyclical, and that she must justify in writing any change in the bank's exposure.

The bank's policy is to compute the Altman Z-score at every review: **Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**, where X1 = working capital ÷ total assets, X2 = retained earnings ÷ total assets, X3 = EBIT ÷ total assets, X4 = market value of equity ÷ total liabilities and X5 = sales ÷ total assets. A score above 2.99 is *safe*, 1.81 to 2.99 *grey* and below 1.81 *distress*. A move from one zone to another is referred to the credit committee, which will want to know not just the score but the reasons for any change. Last year the company scored comfortably in the safe zone.

₹ crore (year-end)	FY2025	FY2026
Current assets	520.0	560.0
Current liabilities	300.0	430.0
Total assets	1,000.0	1,150.0
Retained earnings	380.0	395.0
EBIT	150.0	92.0
Market capitalisation	1,100.0	640.0
Total liabilities	520.0	700.0
Sales	1,300.0	1,250.0

Required

1. Compute X1 to X5 and the Z-score for FY2025 and FY2026, and state the zone for each year.
2. Compute the weighted contribution of each term (1.2X1, 1.4X2, 3.3X3, 0.6X4, 1.0X5) and identify which term explains most of the change in Z.
3. Holding everything else constant, by how much would EBIT have to rise to bring FY2026 Z back to 2.99? Is that realistic?
4. State Anita's recommendation to the credit committee in three lines.

Final answer

Measure	FY2025	FY2026
X1	0.2200	0.1130
X2	0.3800	0.3435
X3	0.1500	0.0800
X4	2.1154	0.9143
X5	1.3000	1.0870
Z-score	3.86	2.52
Zone	Safe	Grey

Z fell by 1.34. The market-value term (0.6X4) accounts for 0.72 of the fall (54%). EBIT would have to rise by about ₹165 crore (from ₹92 crore to about ₹257 crore) to restore 2.99, which is unrealistic. Recommend renewal with enhanced monitoring and quarterly reviews.

CASE Z2 · ALTMAN Z-SCORE

When the Share Price Falls Faster than the Business

Marine Chemicals Ltd.

Learning point: Market-value dependence: how much of Z is the share price, the break-even price, and a book-value cross-check (Z″).

Marine Chemicals, a listed maker of industrial solvents, is a long-standing borrower of Coastal Finance Bank. Three weeks ago a gas leak at its main plant caused a shutdown, a regulatory notice and a social-media storm. The share price fell from ₹90 to ₹38 in five trading days. The latest audited balance sheet, which pre-dates the incident, shows no deterioration, and management insists that "the business fundamentals are unchanged". The bank has ₹120 crore outstanding to the company, secured on plant and inventory. A wrongly triggered watch-list entry could damage a fifteen-year relationship; a missed warning could prove costly.

Rahul Iyer, the bank's risk manager, must tell the risk committee tomorrow whether to move the borrower to the watch list. The early-warning model uses the Altman Z-score (**Z = 1.2X1 + 1.4X2 + 3.3X3 + 0.6X4 + 1.0X5**, with X4 = market value of equity ÷ total liabilities; safe above 2.99, distress below 1.81). Rahul wonders how much of any change in Z comes simply from the price and not from the business. He also has a book-value variant that ignores the share price: **Z″ = 6.56X1 + 3.26X2 + 6.72X3 + 1.05X4**, with X4 = book equity ÷ total liabilities (safe above 2.60, distress below 1.10).

₹ crore unless stated	Latest balance sheet
Current assets	640.0
Current liabilities	410.0
Total assets	1,600.0
Retained earnings	520.0
EBIT	190.0
Total liabilities	900.0
Sales	1,500.0
Shares outstanding (crore)	20.0
Share price before / after the incident (₹)	90 / 38

Required

1. Compute the Z-score at ₹90 and at ₹38, and state the zone in each case.
2. Find the lowest share price at which the company still scores *safe*. Can the company reach the distress zone at any share price?
3. Compute Z″ from book values and compare it with the original Z.
4. Advise the committee: should the borrower go on the watch list, and how should the bank use the two scores together?

Final answer

Measure	Result
Z at ₹90 (market cap ₹1,800 cr)	3.16 (Safe)
Z at ₹38 (market cap ₹760 cr)	2.46 (Grey)
Z with market cap = 0	1.96 (Grey)
Lowest price for "safe"	≈ ₹77.5 (market cap ≈ ₹1,550 cr)
Z″ (book values)	3.62 (Safe)

A 58% price fall moves Z from 3.16 to 2.46 with no change in the business; Z cannot fall below 1.96 even at zero market value, so distress is unreachable. Z″ is safe at 3.62. Put the borrower on the watch list for monitoring (market signal), but do not downgrade the credit until the post-incident accounts are seen.

CASE Z3 · ALTMAN Z-SCORE

No Share Price, No Original Z

Gokul Engineering Pvt. Ltd.

Learning point: Choosing the right model: Z′ for private manufacturers, and why balance-sheet fixes move Z′ less than expected.

Gokul Engineering, a family-owned maker of precision castings, has applied to Western Cooperative Bank for a ₹100 crore term loan to modernise its foundry. The promoters hold all the shares and have never raised outside equity, and the largest customer accounts for 30% of sales. The company is unlisted, so its shares have no market price. Priya Nair, the bank's credit officer, therefore applies the private-company version of the Altman model prescribed in the bank's credit manual: **Z′ = 0.717X1 + 0.847X2 + 3.107X3 + 0.420X4 + 0.998X5**, where X4 = book equity ÷ total liabilities and the other ratios are defined as in the original model. Zones: above 2.90 safe; 1.23 to 2.90 grey; below 1.23 distress.

The first score disappoints. The promoters, keen to win the loan, put forward three proposals: **(A)** convert ₹120 crore of existing long-term loans from the family into equity; **(B)** sell idle land and machinery with a book value of ₹80 crore and use the proceeds to repay term loans (these assets earn no profit); **(C)** cut costs to add ₹20 crore to EBIT, with no change in the balance sheet. The bank's policy requires a safe-zone score for unsecured lending, while a grey-zone score calls for extra security and covenants. Priya must advise which proposals the bank should insist on.

₹ crore (year-end)	Gokul Engineering
Current assets	300.0
Current liabilities	260.0
Total assets	800.0
Retained earnings	60.0
EBIT	70.0
Total liabilities	560.0
Book equity (total assets − total liabilities)	240.0
Sales	960.0

Required

1. Explain in two lines why the original Z cannot be used here. Compute Z′ and state the zone.
2. Recompute Z′ under proposals A, B and C separately. Which gives the largest improvement?
3. Compute Z′ if all three are implemented together. Does the company reach the safe zone?
4. What should Priya conclude about the loan application?

Final answer

Measure	Result
Z′ today	1.75 (Grey)
A: debt-to-equity conversion	1.91 (+0.16)
B: asset sale and repayment	1.95 (+0.20)
C: EBIT up by ₹20 crore	1.83 (+0.08)
A + B + C together	2.35 (Grey)

B gives the largest gain, but no proposal, and not even all three together, reaches 2.90. Z′ is grey, not distress. Approve only with covenants and a staged disbursement tied to operating improvement.